

Software Atelier 4: Software Engineering Project (Spring 2024)

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Batch 2 - Part 1, Quality Requirements (v1, 15/70 points)

Notes (Please Read). Batch 2 includes some mandatory reduction of technical debt. This document highlights this aspect of Batch 2. This task should be planned before starting to work on the User Stories of Batch 2, which will be released on **April 8, 2025**.

Batch 1 Status. If you are very far from completing the user stories of Batch 1, you should focus your effort on that first. Only in the case you have some people who are / would be idle, you can start planning the quality issues.



Technical Debt Reduction and Quality Requirements (15 pts)

These requirements concern only the backend component of your project.

As of **April 7**, the MR related to **Sonarqube Configuration will be merged on dev**. This means that your pipeline on **dev** will likely be failing, because of the configured quality gate:

Metric	Condition on Overall Code
Test Coverage	$\geq 60\%$
Duplicated Lines	$\leq 2\%$
Maintainability Rating	$\geq A$
Reliability Rating	$\geq A$
Security Rating	$\geq A$

You will have access to Sonarqube to check the status of your project.

You need to plan a significant set of tasks to reduce the technical debt of your project to reach the required status of the quality gate. The tasks should be finished by **April 17**.

In general, you can assign the whole team to these tasks (standard way), or you can assign a subset of people from the team, that will focus only on TD reduction, putting a lot of trust in them.

Skipping this task

- You may **decide you want to ignore completely quality**. You will get 0 points for this task, and an additional penalty of 15% (a total of around 35% for batch 2), and a 15% penalty for Batch 3.

You need to decide the way you want to skip quality by **April 9, 2025**.

Grading

In general, the grade will be shared for this task. Exceptions will apply to the personal component of the grade for usual reasons (disappearing or particularly bad code).

The grade will take into account the *reduction of your technical debt, mostly independently of your initial status*. You might be in advantage, but you will be graded on how well you improved your system according to the metrics provided by SonarQube. In particular, the grade will take into account many aspects, including:

- how well your team planned the technical debt reduction, *e.g.*, by properly using issues, MRs, and how well you prioritized the tasks;
- how well you reduced the technical debt, *i.e.*, by providing meaningful fixes (*e.g.*, if you miss documentation, you cannot just write random stuff to let Sonarqube pass), or writing tests properly;
- how well you identified false positives.

The leaders **MUST** tag issues and MRs with the dedicated **Type::TD Reduction** label. Everything else will not be considered for this task. As soon as the task is finished, you must warn the instructor and TA via iCorsi in the dedicated entry.

Penalties

Starting from **April 28**, independently of your validation status of Batch 1, and regarding the whole grade for Batch 2:

- You will get a **team grade penalty of 0.5% for every course day where your quality gate fails on dev and for every component of the quality gate that fails (so from 0.5% to 2.5% maximum)**.
- If you work during vacation days, holidays, or weekends (*i.e.*, if there is any commit of any team member pushed on the repository on such days), you will get a **0.25% team grade penalty for any of such days and for every component where your quality gate fails on dev (so from 0.25% to 1.25%)**.
- For every MR merged, such that **dev** has quality gate failing before merging, you will get an additional **0.5% team grade penalty for any component of the quality gate that fails**;
- For every MR merged such that **dev** fails the quality gate after merging (or if it worsens the number of failing conditions, if it was already failing), you will get a **2% team grade penalty**; The leaders will get a personal penalty of **2% as well**.

Bonuses

- If your team never puts the dev branch in a CI failure state from **April 28** until the validation of all the user stories of Batch 2, you will get a 2.5% bonus on your team grade.
- If your team reaches over 90% test coverage at the time of validation and the tests are meaningful and well organized (following what has been explained through the lectures), you will get another 2.5% bonus on your grade.